Day 8 — Cross-cultural: Lo Shu and Dürer

Module: Magic Squares (*Bhadragaṇita*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will (a) identify the Chinese *Lo Shu* and Albrecht Dürer's *Melencolia I* magic squares, (b) note structural similarities and differences across cultures, and (c) begin their project planning.

Materials

- Image of the Lo Shu square (turtle-shell pattern is helpful)
- Image of Dürer's *Melencolia I* (1514) with the 4×4 corner visible
- Project planning sheet (one per pair)

Vocabulary review

- *Lo Shu* — a Chinese 3×3 magic square attested at least from the *Da Dai Liji* (Han dynasty, ~5th–1st c. BCE) and probably earlier.
- *Melencolia I* — a 1514 engraving by the German artist Albrecht Dürer; its upper-right corner contains a famous 4×4 magic square.

Lesson flow

Warm-up (5 min)

- *"Magic squares are not a one-culture phenomenon. Today we look at two famous non-Indian examples and compare."*
- Show both images side by side, briefly. Don't analyse yet.

Core (20 min)

1. **The Lo Shu (China).**

   The 3×3 magic square in Chinese tradition is identical (up to symmetry) to the 3×3 we've been studying:

   ```
   4 | 9 | 2
   ---------
   3 | 5 | 7
   ---------
   8 | 1 | 6
   ```

   Magic constant: 15 (same as our Day-1 square).

   The legend: Emperor Yu (legendary, ~2000 BCE attribution) saw the pattern on the shell of a turtle emerging from the Lo river. The earliest *documented* appearance in a Chinese text is the *Da Dai Liji* (Han dynasty, roughly 5th–1st c. BCE). Treat the legend as a story; the documented record is the historical evidence.

“This is the same square — up to symmetry — as the one we built on Day 1. Two cultures arrived at the same arrangement independently, separated by huge distances.”

2. **Dürer’s *Melencolia I* (Germany, 1514 CE).**

   The 4×4 in Dürer’s engraving:

   ```
   16 |  3 |  2 | 13
   ----------------
    5 | 10 | 11 |  8
   ----------------
    9 |  6 |  7 | 12
   ----------------
    4 | 15 | 14 |  1
   ```

   Magic constant: 34.

   Notable features:

   - The bottom-middle two cells contain 15 and 14 — encoding the date **1514**.
   - The four corners (16, 13, 4, 1) sum to 34.
   - Each 2×2 panel sums to 34 (panel-magic property — same as Khajuraho!).
   - Dürer’s square is not the same as Khajuraho’s, but shares the panel-magic property.

3. **Compare and contrast.** Build a class table:

Square	Approx. date	Culture	Order	Magic constant	Special features
Lo Shu	~5th–1st c. BCE (documented)	Chinese	3×3	15	Identical to our 3×3
Khajuraho	~10th–11th c. CE	Indian	4×4	34	Panel-magic
Nārāyaṇa’s worked examples	1356 CE	Indian	up to large *n*	varies	Systematic methods
Dürer	1514 CE	German	4×4	34	Panel-magic; encodes year

4. **Honest framing.** Magic squares appeared independently (or via transmission) in:

   - **China:** Lo Shu and later expansions.
   - **India:** numerous artefacts and Nārāyaṇa’s systematic methods.
   - **Islamic world:** treatises by al-Bīrūnī (~11th c. CE) and others.

    - **Europe:** through Islamic and (likely) Indian transmission by 14th–15th c.

We do not have a clean story of “X invented it; Y borrowed it.” Frame the topic as a *cross-cultural mathematical fascination*.

Activity (15 min) — Project planning

- Use `activities/activity-02-cross-cultural-compare.md` (a side-by-side worksheet).
- Pairs commit to their project format (poster / video / written exposition) and one or two key ideas.
- Each pair writes a 3-sentence project plan, hands in for teacher review by end of class.

Wrap-up (5 min)

- Two-pair quick share: “Our project is going to be ___ because ___.”
- Preview Day 9: project work + a brief look at Sudoku.

Student-facing content

Magic squares across cultures.

```
Lo Shu (China, ~5th c. BCE)
4 | 9 | 2
---------
3 | 5 | 7
---------
8 | 1 | 6
```

```
 Dürer (Germany, 1514 CE)
16 |  3 |  2 | 13
----------------
 5 | 10 | 11 |  8
----------------
 9 |  6 |  7 | 12
----------------
 4 | 15 | 14 |  1
```

The Lo Shu 3×3 is the same square as our Day-1 example (up to symmetry). The Dürer 4×4 shares the **panel-magic** property with the Khajuraho square (~10th–11th c. CE) — every 2×2 panel sums to 34.

Multiple cultures arrived at magic squares: China (~5th c. BCE earliest documented), India (Khajuraho ~10th c. CE, Nārāyaṇa’s *Bhadragaṇita* 1356 CE), the Islamic world (~11th c. CE on), and Europe (~14th c. CE on). No clean single origin.

Homework

- Choose one of the three squares we’ve seen this fortnight (Day-1 / Khajuraho / Dürer). Verify its magic constant on a fresh piece of paper from scratch. Note one feature you find interesting.

Differentiation

- **One tier up:** Look up one more cultural example (e.g. al-Bīrūnī’s magic squares, or the Jain *Pārśvanātha* yantra). Bring an image and a verified magic constant.
- **One tier down:** Focus on one comparison: how is the Lo Shu the same as our Day-1 square? Trace the rotation that takes one into the other.

Teacher notes

- Don’t fall into “ancient Indians invented this first” framing. The Lo Shu is older than any dated Indian magic square we have. Nārāyaṇa’s contribution is the *systematic method*, not the *first discovery*.
- The Dürer corner being a date (1514) often delights students. Use that delight to motivate the project.
- Some students will see the Khajuraho ~10th-c. dating and conclude it predates Lo Shu. That is incorrect. The Lo Shu has earlier documented evidence (Han dynasty texts). Be clear about which dates are *attestations* and which are *legends*.

Citation(s) used in this lesson

- Lo Shu: *Da Dai Liji* (Chinese, Han dynasty, ~5th–1st c. BCE) — earliest documented Chinese mention.
- Dürer, *Melencolia I* (1514) — material artefact.
- Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī*, Book 14, 1356 CE — for context on Indian systematic treatment.